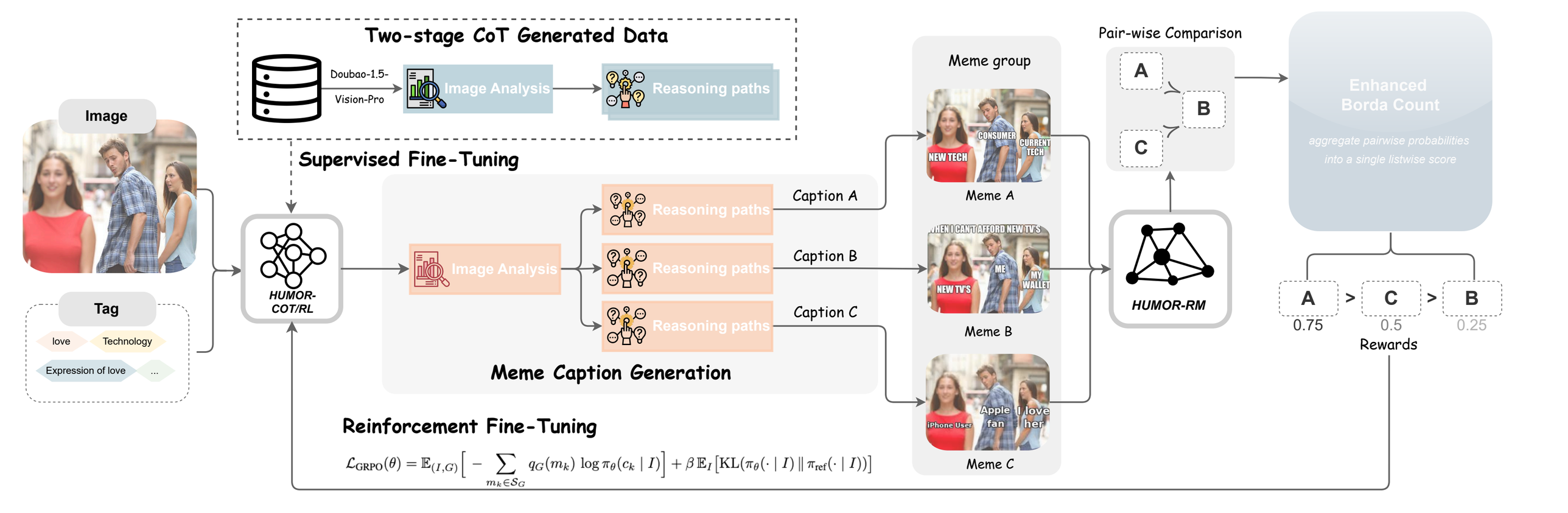
Two-stage CoT Generated Data
Doubao-1.5-Vision-Pro
Image Analysis
Reasoning paths
Supervised Fine-Tuning
Image
Tag
love
Technology
Expression of love
...
HUMOR-COT/RL
Image Analysis
Reasoning paths
Reasoning paths
Reasoning paths
Caption A
Caption B
Caption C
Meme Caption Generation
Meme group
NEW TECH
CONSUMER
CURRENT TECH
Meme A
WHEN I CAN'T AFFORD NEW TV'S
NEW TV'S
ME
MY WALLET
Meme B
iPhone User
Apple fan
I love her
Meme C
Pair-wise Comparison
A
B
C
HUMOR-RM
Enhanced Borda Count
aggregate pairwise probabilities into a single listwise score
A > C > B
0.75
0.5
0.25
Rewards
Reinforcement Fine-Tuning
$\mathcal{L}_{\text{GRPO}}(\theta) = \mathbb{E}_{(I,G)}\Big[- \sum_{m_k \in \mathcal{S}_G} q_G(m_k)\, \log \pi_\theta(c_k \mid I)\Big] + \beta\, \mathbb{E}_I\big[\mathrm{KL}(\pi_\theta(\cdot \mid I) \,\|\, \pi_{\text{ref}}(\cdot \mid I))\big]$